Name: _____

Student Workbook *w/Solutions*

Fall, 2022 - Codap Edition

Workbook v1.5

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Introduction to Computational Data Science

Many important questions ("What's the best restaurant in town?", "Is this law good for citizens?", etc.) are answered with *data*. Data Scientists try and answer these questions by writing *programs that ask questions about data*.

Data of all types can be organized into **Tables**.

- Every Table has a **header row** and some number of **data rows**.
- **Quantitative data** is numeric and measures *an amount*, such as a person's height, a score on a test, distance, etc. A list of quantitative data can be ordered from smallest to largest.
- **Categorical data** is data that specifies *qualities*, such as sex, eye color, country of origin, etc. Categorical data is not subject to the laws of arithmetic — for example, we cannot take the "average" of a list of colors.

The Animals Dataset

name	species	sex	age	fixed	legs	pounds	weeks
Sasha	cat	female	1	false	4	6.5	3
Snuffles	rabbit	female	3	true	4	3.5	8
Mittens	cat	female	2	true	4	7.4	1
Sunflower	cat	female	5	true	4	8.1	6
Felix	cat	male	16	true	4	9.2	5
Sheba	cat	female	7	true	4	8.4	6
Billie	snail	hermaphrodite	0.5	false	0	0.1	3
Snowcone	cat	female	2	true	4	6.5	5
Wade	cat	male	1	false	4	3.2	1
Hercules	cat	male	3	false	4	13.4	2
Toggle	dog	female	3	true	4	48	1
Boo-boo	dog	male	11	true	4	123	24
Fritz	dog	male	4	true	4	92	3
Midnight	dog	female	5	false	4	112	4
Rex	dog	male	1	false	4	28.9	9
Gir	dog	male	8	false	4	88	5
Max	dog	male	3	false	4	52.8	8
Nori	dog	female	3	true	4	35.3	1
Mr. Peanutbutter	dog	male	10	false	4	161	6
Lucky	dog	male	3	true	3	45.4	9
Kujo	dog	male	8	false	4	172	30
Buddy	lizard	male	2	false	4	0.3	3
Gila	lizard	female	3	true	4	1.2	4
Bo	dog	male	8	true	4	76.1	10
Nibblet	rabbit	male	6	false	4	4.3	2
Snuggles	tarantula	female	2	false	8	0.1	1
Daisy	dog	female	5	true	4	68	8
Ada	dog	female	2	true	4	32	3
Miaulis	cat	male	7	false	4	8.8	4
Heathcliff	cat	male	1	true	4	2.1	2
Tinkles	cat	female	1	true	4	1.7	3
Maple	dog	female	3	true	4	51.6	4

Categorical or Quantitative?

For each piece of data below, circle whether it is **Categorical** or **Quantitative** data.

1	Hair color	**CATEGORICAL**	quantitative
2	Age	categorical	**QUANTITATIVE**
3	ZIP Code	**CATEGORICAL**	quantitative
4	Date	**CATEGORICAL**	quantitative
5	Height	categorical	**QUANTITATIVE**
6	Sex	**CATEGORICAL**	quantitative
7	Street Name	**CATEGORICAL**	quantitative

For each question, circle whether it will be answered by **Categorical** or **Quantitative** data.

8	We'd like to find out the average price of cars in a lot.	categorical	**QUANTITATIVE**
9	We'd like to find out the most popular color for cars.	**CATEGORICAL**	quantitative
10	We'd like to find out which puppy is the youngest.	categorical	**QUANTITATIVE**
11	We'd like to find out which cats have been fixed.	**CATEGORICAL**	quantitative
12	We want to know which people have a ZIP code of 02907.	**CATEGORICAL**	quantitative
13	We'd like to sort a list of phone numbers by area code.	categorical	**QUANTITATIVE**

Questions and Column Descriptions

What questions can you ask about the animals dataset? For each question, **can it be answered by this dataset?** Make sure you have at least two questions that can be answered, and at least one that cannot.

What do you NOTICE about this dataset?	What do you WONDER about this dataset?	Answered by this dataset?
Not every animal has the same number of legs	*Why does Lucky only have 3 legs?*	*Yes No*
		Yes No
		Yes No
		Yes No
		Yes No
		Yes No
		Yes No

Describe the table, and two of the columns, by filling in the blanks below.

1. This dataset is *Animals that came from an animal shelter*, which contains *32* data rows.
2. Some of the columns are:
   a. *species* (column name), which contains *categorical* (categorical or quantitative) data. Some example values are: *"cat", "dog", and "rabbit"*.
   b. *age* (column name), which contains *quantitative* (categorical or quantitative) data. Some example values are: `1`, `5`, *and* `10`.

What's on your mind?

Simple Data Types

CODAP is a web-based data science tool that runs in your web browser.

Data Types

CODAP utilizes different ***data types***, including Numbers, Strings, and Booleans.

- Numbers are values like `1`, `0.4`, `1/3`, and `-8261.003`.
  - Numbers are *usually* used for quantitative data and other values are *usually* used as categorical data.
- Strings are values like `"Emma"`, `"Rosanna"`, `"Jen and Ed"`, or even `"08/28/1980"`.
  - All strings *must* be surrounded in quotation marks.
- Booleans are either `true` or `false`.

All values evaluate to themselves. The program `42` will evaluate to `42`, the String `"Hello"` will evaluate to `"Hello"`, and the Boolean `false` will evaluate to `false`.

Operators

Operators (like `+`, `-`, `<`, etc.) work the same way that they do in math.

- Operators are written between values, for example: `4 + 2`.

Expressions

Expressions work the same way that they do in math. Numeric expressions can be *evaluated*.

- The following are all examples of expressions: `sqrt(16)`, `sqrt(Weight)`, `m+5`, and `9+17`.

CODAP Exploration

Make sure you're logged into the Animals Starter File in CODAP. The purpose of this exploration is to familiarize yourself with some of CODAP's features. Feel free to experiment and try things out!

1) See if you can make a graph. What did you select in order to do so?

I clicked on the "Graph" icon in the upper left-hand corner of my screen

2) Move the table. Move the graph. How do you move things in CODAP?

To move tables and graphs, I mouse over the top-most bar until my cursor turns into a hand. Then I can drag.

3) Minimize and maximize the table and the graph. How do you do that?

To minimize, I select the `–` *in the upper right-hand corner of the graph or table. To maximize, I reselect the* `–`.

4) Drag an attribute (like `weight`, `name`, or `sex`) to a graph's axis - or use the drop-down menu that appears when you click on an axis. What happens?

The randomly positioned points move to the confirguration that matches the data on the animals table.

5) Drag a second attribute (like `weight`, `name`, or `sex`) to the graph's other axis - or use the drop-down menu that appears when you click on an axis. What happens?

The points move again, to the confirguration indicated by the two attributes on the animals table.

6) Select the grey plus sign (+) in the upper right-hand corner of the Animals Table. Doing so will enable you to create a new attribute. Name your new attribute `test`. Is the column populated or empty?

The column is empty.

Numbers and Strings

Make sure you're logged into CODAP, and have created a new column titled `test` in the Animals Dataset. In order to follow each of these directives, you must first select `test`, and then choose `Edit formula` from the drop-down menu that appears. You will be typing into the `Edit Formula` box.

Numbers

1) Try typing `42` and then select `Apply`. What happens? What is the largest number CODAP can handle?

I didn't find a number that was too big!

2) What happens if you try a fraction like `1/5`? How about `1/3`?

The editor will return the decimal equivalent. Note: The precision for decimals is set to two place values. So, `1/3` *will evaluate to* `0.33`.

Students may edit attribute properties and adjust the precision up to 8 digits.

3) Try writing negative integers, fractions and decimals.

Strings

String values are always in quotes.

4) Try typing your name in the `Formula` box. What error message appears? Why do you think it appears?

5) Try typing your name in quotes.

6) Try typing your name with the opening quote, but *without the closing quote.* Read the error message!

7) Is `42` the same as `"42"`? Why or why not? Write your answer below:

No. CODAP always interprets 42 as a number.

8) Explain what you understand about how strings work in CODAP.

Strings require quotes.

Operators

9) Just like math, CODAP has ***operators*** like `+`, `-`, `*` and `/` and `^`. Try typing in `4 + 2`.

10) Type in and apply the following expressions, **one at a time**: `4 + 2 * 6`, `(4 + 2) * 6`, `4 + (2 * 6)`. What do you notice? Does CODAP know the order of operations?

Yes, CODAP automatically applies the order of operations.

11) Try typing in `4 + "cat"`, and then `"dog" + "cat"`. What can you conclude from this?

You can use `+` *to combine two strings, and you can even combine a number and a string!*

BUT, if you add 4 + "6", CODAP will convert the "6" into a Number and produce `10`*!*

Booleans

Boolean-producing expressions are yes-or-no questions and will always evaluate to either `true` ("yes") or `false` ("no"). What will each of the expressions below evaluate to? **In order to try out the code, you must first select `test`, and then choose `Edit formula` from the drop-down menu that appears.** You will be typing into the `Edit Formula` box.

	Prediction:	Computer Returns:		Prediction:	Computer Returns:
1) `3 <= 4`		*true*	2) `"a" > "b"`		*false*
3) `3 = 2`		*false*	4) `"a" < "b"`		*true*
5) `2 < 4`		*true*	6) `"a" = "b"`		*false*
7) `5 >= 5`		*true*	8) `"a" != "a"`		*false*
9) `4 >= 6`		*false*	10) `"a" >= "a"`		*true*
11) `3 != 3`		*false*	12) `"a" != "b"`		*true*

13) In your own words, describe what `<` does. *checks whether the first number is smaller than the second number, returns true if it is and false if it isn*

14) In your own words, describe what `>=` does. *checks whether the first number is greater than or equal to the second number, returns true if it is and*

15) In your own words, describe what `!=` does. *checks whether the to numbers are unequal, returns true if they're unequal and false if they're equal*

	Prediction:	Computer Returns:
16) `includes("cat", "catnap")`		*true*
17) `includes("catnap", "cat")`		*false*

18) How many **Numbers** are there in the entire universe? *infinite*

19) How many **Strings** are there in the entire universe? *infinite*

20) How many **Booleans** are there in the entire universe? *two*

Expressions & Functions in CODAP

Complete the tables below.

$y = 3x$			$y = x + 2$	
x	y		x	y
5	*15*		3	*5*
3.5	*10.5*		8	*10*
7.1	*21.3*		1	*3*
8.4	*25.2*		6	*8*
9.2	*27.6*		5	*7*

1) Open your Animals Starter File. Edit the `test` column's formula to compute *triple* the weight of all of the animals. What formula will you use? *Note: Attribute names that are more than one word need to be entered inside of tick marks.*

*`Weight (lbs)` * 3*

- Compare your results with the first table, above.

2) Edit the `test` column's formula so that 2 weeks are added to the time to adoption for all of the animals. What formula did you use?

`Time to adoption (weeks)` + 2

Applying Functions

In order to answer each of the questions below, you must first select `test` in Animals Starter File, and then choose `Edit formula` from the drop-down menu that appears. You will be typing into the `Edit Formula` box.

- Enter `sqrt(16)` into the `Edit Formula` box.

3) How many arguments does `sqrt` expect? *1*

4) What type of argument does the function expect? (Number? String? Boolean?) *Number.*

5) What type of output does `sqrt` produce? *Number.*

6) What did the function evaluate to? *4*

- Now, enter `sqrt(Weight)`.

7) What do you observe? *CODAP finds the square root of each animal's weight*

8) What happens when you enter `sqrt(Legs)`? *CODAP finds the square root of each animal's legs for* `sqrt(Legs)`.

9) What about `sqrt(Name)`? *CODAP produces an error message because it cannot find the square root of a string, only a number.*

- Enter `stringLength(Name)` into the `test` column.

10) What do you observe? *7 appears, CODAP finds the amount of letters in each animal's name*

11) How many arguments does `stringLength` expect? *1*

12) What type of argument does `stringLength` expect? *String*

13) Does `stringLength` produce a Number? String? Boolean? *Number*

Putting it all together

- Create a column called `age-next-year` that uses an expression to add one year to each animal's age.
- Create a column called `rounded-weight` that uses a function to give the weight of each animal rounded to the nearest whole number.
- Create a column called `old` that uses a Boolean to indicate which animals are over 5 years old.

Displaying Categorical Data

Displaying Categorical Variables

- With a table open in CODAP, select the "graph" icon to produce a scatter plot of *randomly distributed* data points.
- Drag attributes/columns to the axes (or select from a drop-down menu of attributes/columns by clicking the axes) to organize the data so that it is no longer randomly distributed.
- Once the data is organized, manipulate it further by selecting the graph menu icons:
  - **the ruler icon** provides options for calculating statistics such as mean, median, and standard deviation
  - for *some* datasets (those with strong correlations), the ruler icon will offer *additional* statistical computations (such as a least squares line or regression line)
  - the **bar graph icon** allows new configurations of the data. Select this option to group data points into bins or create a bar for each point. Clicking on the bar graph icon a second time (for instance, after data is grouped into bins) allows the creation of a histogram (by fusing the dots into bars).

Exploring other Displays

Data Scientists use **displays** to visualize data. You've probably seen some of these charts, graphs and plots yourselves! When it comes to displaying **Categorical Data**, we often rely on **bar charts** and **pie charts**. CODAP, however, only has the capacity to create **bar charts**.

When we want to create a data display in CODAP, it is important to consider the following: Which attributes on which axes? What type of data? What configuration?

Bar charts show the *count or percentage* of rows in each category.

- Bar charts provide a visual representation of the frequency of values in a categorical column.
- Bar charts have a bar for every category in a column.
- The more rows in a category, the taller the bar.
- Bars in a bar chart can be show in *any order*, without changing the meaning of the chart. However, bars are usually shown in some sensible order (bars for the number of orders for different t-shirt sizes might be presented in order of smallest to largest shirt).

Data Displays in CODAP

Make sure you're logged into CODAP, with the Animals Dataset open. Create a graph of randomly generated points by selecting the `Graph` icon, and then respond to the following prompts.

Creating Bar Charts in CODAP

1) Select the y-axis on your graph (where it says "Click here"). On the drop-down menu that apperas, select `fixed`. (*If you prefer, you may also drag the attribute name from the table to the y-axis.*) What do you notice?

A dot plot forms, categorizing animals as fixed (true) and not fixed (false). The labels for the dot plot are on the y-axis. The dots are stacked horizontally.

2) Now select the x-axis on your graph and select `fixed`. How does the graph change?

The labels move from the y-axis to the x-axis, and the dots are stacked vertically.

3) Select the `configuration` icon (which looks like a bar graph) to the right of the data display. Select `fuse dots into bars`

4) Click the `ruler` icon, and test out each of the two options available (count and percentage). What happens?

When I select "count," I see the number of animals represented by each bar. When I select "percentage," the percentage of animals represented by each bar appears.

5) Now, make a bar chart showing how many animals there are of each species by changing the variable on the x-axis to `species`. Can you figure out how to reconfigure the bar chart as a dot plot?

To reconfigure as a dot plot, select the `configuration` *icon and deselect* `fuse dots into bars`.

Numeric versus Categorical Displays

6) Create a graph with `weight` on the x-axis. Then, click on `weight` so that a drop-down menu appears. From this drop-down menu, choose `Treat as Categorical`, and observe how the graph changes. What is going on?

CODAP is no longer interpreting weight as numerical data. Rather, creating many categories; each weight category contains just 1 or 2 animals.

7) Can you `fuse dots into bars` when treating `weight` categorically? Numerically? Why does CODAP *sometimes* invite you to `fuse dots into bars`, but other times the option does not appear?

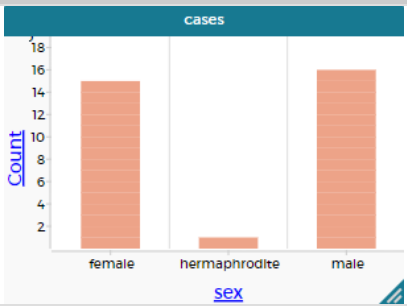
Bar charts show categories, therefore are conducive to representing categorical data. They classify, rathern than measure

Practice Plotting

Fill in the tables, predict what the graph will look like (a rough sketch is fine!), and then use CODAP to make the following displays.

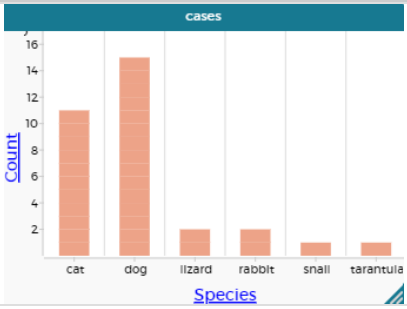
1) A bar chart showing the `sex` of animals from the shelter.

Column / Attribute	Type of Data	Configuration
sex [column used as x-axis] *n/a* [column used as y-axis]	☐ *Quantitative* ✓ *Categorical*	☐ *Points* ✓ *Fuse dots into bars* ☐ *Bar for each point* ☐ *Group into bins*

Sketch the chart below:	What do you think the display tells us?
	*The display tells us how many animals are male, hermaphrodite, and female.*

2) A bar chart showing the `species` of animals from the shelter.

Column / Attribute	Type of Data	Configuration
species [column used as x-axis] *n/a* [column used as y-axis]	☐ *Quantitative* ✓ *Categorical*	☐ *Points* ✓ *Fuse dots into bars* ☐ *Bar for each point* ☐ *Group into bins*

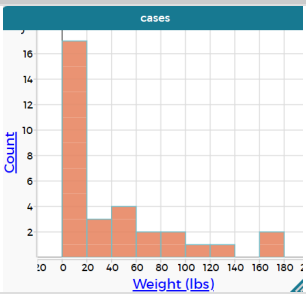
Sketch the chart below:	What do you think the display tells us?
	*The display tells us how many animals are cats, dogs, lizards, rabbits, snails, and tarantulas.*

Practice Plotting 2

Fill in the tables, predict what the graph will look like (a rough sketch is fine!), and then use CODAP to make the following displays.

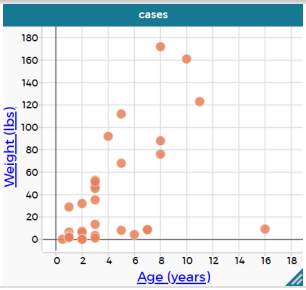
1) A histogram of the number of `pounds` that animals weigh.

Column / Attribute	Type of Data	Configuration
pounds [column used as x-axis] *n/a* [column used as y-axis]	✓ *Quantitative* ☐ *Categorical*	☐ *Points* ✓ *Fuse dots into bars* ☐ *Bar for each point* ✓ *Group into bins*

Sketch the chart below:	What do you think the data display tells us?
	*The display shows us the distribution of animals' weights into varying increments.*

2) A scatter-plot, using the animals `name` as the labels, `age` as the x-axis, and `pounds` as the y-axis, for all the animals from the shelter.

Column / Attribute	Type of Data	Measure
age *[column used as x-axis]* *pounds* *[column used as y-axis]* *name* *[(optional) column used for labels]*	✓ *Quantitative* ☐ *Categorical*	☐ *Box plot* ☐ *Least squares line* ✓ *No need to make a selection*

Sketch the chart below:	What do you think the data display tells us?
	*The display shows us the relationship between animals' age and weight.*

Practice Plotting - Challenge

Fill in the tables, predict what the graph will look like (a rough sketch is fine!), and then use CODAP to make the following displays.

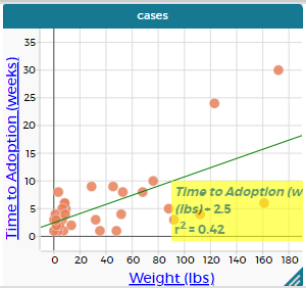
1) A scatter-plot, using the animals `species` as the labels, `age` as the x-axis, and `weeks` as the y-axis, for all the animals from the shelter.

Column / Attribute	Type of Data	Measure
age [column used as x-axis] *weeks* [column used as y-axis] *species* [(optional) column used for labels]	✓ *Quantitative* ☐ *Categorical*	☐ *Box plot* ☐ *Least squares line* ✓ *No need to make a selection*

Sketch the chart below:	What do you think the data display tells us?
scatter plot showing animals' age compared to time to adopt	*The display shows us the relationship between animals' age and weight.*

2) (Challenge) A least squares line, using the animals `species` as the labels, `pounds` as the x-axis, and `weeks` as the y-axis, for all the animals from the shelter.

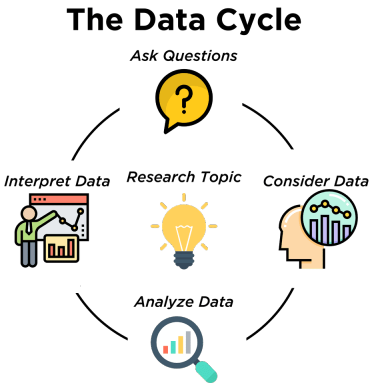
Column / Attribute	Type of Data	Measure
pounds [column used as x-axis] *weeks* [column used as y-axis] *species* [(optional) column used for labels]	✓ *Quantitative* ☐ *Categorical*	☐ *Box plot* ✓ *Least squares line* ☐ *No need to make a selection*

Sketch the chart below:	What do you think the data display tells us?
	*The display shows us the relationship between animals' weight and weeks to adoption.*

What's on your mind?

The Data Cycle

Data Science is all about *asking questions of data*. Sometimes the answer is easy to compute. Sometimes the answer to a question is *already in the dataset* - no computation needed. And sometimes the answer just sparks more questions! Data Scientists ask a lot of questions, and each one adds a chapter to the story of their research. Even if a question is a "dead-end", it's valuable to share what the question was and what work you did to answer it!



We start by **Asking Questions** - questions that can be answered with data. It's useful to think of all questions as falling into four categories:

- **Lookup questions** can be answered simply by looking up a single value in the table. Examples of lookup questions might be "How many legs does Felix have?" or "What species is Sheba?"
- **Arithmetic questions** can be answered by computing an answer across a single column. Examples of arithemetic questions might be "How much does the heaviest animal weigh?" or "What is the average age of animals from the shelter?"
- **Statistical questions** are where things get interesting! The answers are never black-and-white. If we asked, "How old are animals at the shelter?" we could report back the average age, the age that shows up most frequently or the range of the ages. There are also some statistical questions that deal with *relationships between* ***two*** *columns*: "Do cats tend to be adopted faster than dogs?" or "Are older animals heavier than young ones?"
- **Questions we can't answer** are pretty common, too! We might wonder where the animal shelter is located, or what time of year the data was gathered! But the data in the table won't help us answer that question, so as Data Scientists we might need to do some research *beyond the data*. And if nothing turns up, we simply recognize that there are limits to what we can analyze.

Next, we **Consider Data**. This could be done by conducting a survey, observing and recording data, or finding a dataset that meets our needs. Since our data is contained in a table, it's useful to start by asking two questions:

- What rows do we care about? - Is it all the animals? Just the lizards?
- What columns do we need? - Are we examining the ages of the animals? Their weights?

Then, we **Analyze the Data**, by producing data displays and new tables of filtered or transformed data and using them to identify patterns and relationships.

- Are we making a pie chart? A bar chart? Something else?

Finally, we **Interpret the Data**, by answering our questions and summarizing the results. As we've already seen from the Animals Dataset, these interpretations often lead to *new* questions… and the cycle begins again.

Which Question Type?

Take a moment to look at the table of Pokemon data below:

name	type1	hitpoint	attack	defense	speed
Bulbasaur	Grass	45	49	49	45
Ivysaur	Grass	60	62	63	60
Venusaur	Grass	80	82	83	80
Mega Venusaur	Grass	80	100	123	80
Charmander	Fire	39	52	43	65
Charmeleon	Fire	58	64	58	80
Charizard	Fire	78	84	78	100
Mega Charizard X	Fire	78	130	111	100
Mega Charizard Y	Fire	78	104	78	100
Squirtle	Water	44	48	65	43
Wartortle	Water	59	63	80	58

For each question about this data below, write down whether it's a Lookup, Arithmetic, or Statistical Question.

	Question	Type	Which Rows?	Column(s)?
1	What type is Charizard?	*Lookup*	*Charizard*	*Type*
2	Which Pokemon is the fastest?	*Arithmetic*	*All rows*	*Speed*
3	What is Wartortle's attack score?	*Lookup*	*Wortotle*	*Attack*
4	What is the average defense score?	*Arithmetic*	*All rows*	*Defense*
5	What is a typical defense score?	*Statistical*	*All rows*	*Defense*
6	Is Ivy faster than Venusaur?	*Arithmetic*	*Ivy and Venusaur*	*Speed*
7	Does speed affect attack score?	*Statistical*	*All rows*	*Speed and Attack*
8	What is the most common type?	*Arithmetic*	*All rows*	*Type*
9	Does one type tend to be faster than others?	*Statistical*	*All rows*	*Type and Speed*
10	Are hit-points (hp) similar for all Pokemon?	*Statistical*	*All rows*	*hp*
11	Which Pokemon has a speed of 43?	*Lookup*	*Squirtle*	*Name*

Data Cycle: Consider Data

For each question about the animals dataset below, identify the type of question and fill in the Rows and Columns needed to answer it.

Ask Questions

How old is Boo-boo?
What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

just Boo-Boo
What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

age
What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Ask Questions

Are there more cats than dogs in the shelter?
What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

cats and dogs
What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

any column / all columns
What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Ask Questions

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

cats and dogs
What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

any column / all columns
What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Ask Questions

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

cats and dogs
What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

any column / all columns
What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Data Cycle: Analyzing with Displays

For each question below, complete the first three steps of the Data Cycle. Once you know what code to write, type it into Pyret and try it out!

Ask Questions	*How many of each species are at the shelter?* *What question do you have?* What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	*What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	

Ask Questions	*How many of each sex are at the shelter?* *What question do you have?* What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	*What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	

Ask Questions	*What question do you have?* What type of question is this? (circle one) Lookup Arithmetic Statistical
Consider Data	*What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)* *What Columns do we need? (age, weight-in-kilograms, weeks, etc.)*
Analyze Data	

Using Transformers: Sort, Filter, Transform Attribute, Build Attribute

Transformers allow us to transform datasets to produce new, distinct output datasets, instead of modifying the original input dataset itself. We use them to manipulate tables and enable low-stakes "what if?" exploration.

- In this course, we'll be using a variety of Transformers.
  - `Sort` - We must provide `Sort` with a dataset, a sort method (we'll focus on sorting by attribute in this course), and a sort direction. `Sort` produces a copy of the input dataset, with cases sorted by the selected attribute.
  - `Filter` - We must provide `Filter` with an expression that evaluates to either true or false. `Filter` then produces a copy of the input dataset that only has the cases for which the formula was true.
  - `Transform Attribute` - We provide `Transform Attribute` with an attribute to transform, a new name for the transformed attribute, and an expression to apply to each row. The `Transform Attribute` replaces the value of the designated attribute with the result of the expression. `Transform Attribute` produces a copy of the input dataset with the designated row transformed.
  - `Build Attribute` - We provide an expression that is used to calculate an attribute's new values. `Build Attribute` produces a copy of the input dataset with the new attribute added.
- With the exception of `Sort`, each of the above Transformers is attached to a unique expression. It's important to get the expression just right, or the Transformer will produce an error. Strings belong inside quotation marks, but Booleans do not! Attribute names that include units (for instance, `Age (years)`) belong inside tick marks (`). }

Sort Transformer

Make sure you're logged into CODAP with the Animals Dataset open. Open the Transformers plugin, and select `Sort`.

Creating a Sort Transformer

- Select `Sort` from the drop-down menu of Transformers.
- We want to give each Transformer a useful name, one which clearly describes what that Transformer does. Generally, a brief hyphenated description works - for instance: `order-by-pounds`.

1) What would be a good name for a Transformer that organizes the table alphabetically by animals' names?

order-by-name.

- Type in the name you wrote above.

2) Which dataset do you want CODAP to apply this Transformer to? *Animals Dataset*

- Select the appropriate dataset from the drop-down menu.
- Select `Name` as the attribute to sort.

3) Which direction will you select? Why?

"Ascending" sorts from A to Z, so I will select that.

Applying and Saving the Sort Transformer

4) Apply the Transformer. What happens?

A copy of the original dataset is created.

5) What do you notice about the name of the new dataset?

It has the Transformer's customized name at the front, with the name of the original dataset in parentheses.

6) What happens when you save a Transformer?

When I save a Transformer, a small box appears that has my Transformer's name at the top. I will now be able to use the Transformer again, without inputting all of the different specifications.

7) How is saving a Transformer different from saving a dataset?

Saving a dataset saves the table, but not the operations performed on it. Saved tables appear on the drop-down menu of datasets to transform. Saving a transformer lets us re-use an operation.

Filter

Make sure you're logged into the Animals Starter File in CODAP.

Creating a "Filter" Transformer

Open the Transformers plugin, and select `Filter`.

1) Enter a `filter-is-fixed` as the name. What does this name tell you about the Transformer you are creating?

The Transformer will filter out only animals that are fixed.

2) Which dataset should we select, to apply this Transformer? *Animals Dataset*

3) Examine the contract. Why does the provided contract make sense?

The expression will consume a Row and produce a Boolean.

4) Write a purpose statement.

The expression will consume a Row and produce a Boolean.

- Enter `Fixed = true` as the expression.

5) Select `Apply Transformer`. What happens?

A copy of the original dataset is created, with only animals that are fixed.

6) Try typing `fixed = true` as the expression (instead of `Fixed = true`). What happens?

An error message appears that says, "Evaluating `fixed = true` failed: 'fixed' is unrecognized."

7) What happens when you enter `FIXED = true`? How about `Fixed = TRUE`?

Error messages appear stating that FIXED and TRUE are unrecognized.

More Filtering

8) Create, save, and apply a Transformer called `filter-is-old` that creates a new dataset with animals older than 5 years. Note that `Age (years)` needs to be typed within backticks``` in order for CODAP to recognize it as an attribute. (The backtick key is in the upper left corner of the keyboard.) How many rows does the resulting table have?

There are 9 rows.

9) Create, save, and apply a Transformer called `filter-is-dog` that creates a new dataset with only dogs. How many dogs are there at the shelter?

There are 15 dogs.

Transform

Make sure you're logged into CODAP with the Animals Dataset open.

Creating a "Transform" Transformer

- Select `Transform Attribute` in the Transformer plugin.
- Enter `transform-is-young` as the name. This Transformer will replace all ages less than 5 with the Boolean `true`. In other words, it will *transform* our "age" column into a column that tells us if an animal is young or not.
- Select the Animals Dataset.

1) What would be an appropriate name for our transformed attribute? `Young`

2) Examine the contract for the expresson. A Domain (row) is provided… but not a Range. What type of data do we expect for this expression?

3) Let's write a purpose statement. What do you think we want a Transformer called `transform-is-young` to do?

Creates a new table where ages below 5 are replaced with an indication of whether the animal is young or not.

- Enter `Age (years) < 5` as the expression. (Include tick marks around `Age (years)`.)
- Apply the Transformer. To define the Transformer for future use, select `Save`.

More Transforming

4) Now, try creating a Transformer called `weight-in-kg`. (Note: To convert pounds to kilograms, divide pounds by 2.205.) How many kilograms is the heaviest animal in the shelter?

Kujo, the heaviest animal, is 78 kilograms.

Build Attribute

Make sure you're logged into CODAP with the Animals Dataset open.

Creating a Build Attribute Transformer

1) Now select `Build Attribute` in the Transformer plugin.

2) Enter `build-is-heavy` as the name. What does this name tell you about the Transformer we are creating?

We are going to build a column that states `true` *for heavy animals and* `false` *for lightweight animals.*

3) Select `animals-dataset`.

4) What would be an appropriate name for our *new* attribute?

`Heavy`

5) Examine the contract. A domain is provided (row), but not a range. What is the desired output for `build-is-heavy`?

Boolean.

6) Let's write a purpose statement. What do you think we want a Transformer called `build-is-heavy` to do?

`build-is-heavy` *consumes a weight and returns* `true` *if the weight is greater than a specified amount.*

7) Enter `Weight > 40` as the expression.

8) Apply the Transformer. To define the Transformer for future use, select `Save`.

More Building

9) Now, try creating a Transformer called `updated-age`. This Transformer will give the animals' ages one year from today. How many animals are 9 years-old one year from today?

In one year, there are three animals that are 9 years-old.

Transformer Chaining

Transformer chaining allows us to apply multiple Transformers with fewer clicks.

Let's say we create the following Transformers:

- `build-in-kgs` - builds a new column with pounds converted to kilograms
- `filter-is-lightweight` - keeps all rows where weight is no more than 9 kilograms
- `sort-ascending` - sorts the dataset from lightest to heaviest weight

When we *save* Transformers, we can apply them to new datasets - preventing lots of duplicate work. Additionally, when we have a chain of transformed datasets and change the original dataset, the updates will flow through the chain.

Order Matters! Transformers must be applied in the appropriate order given the dataset. For example, trying to order a table by a column that hasn't been built will result in an error.

Design Recipe

Directions: Define a Transformer called `filter-is-dog`, whose expression consumes a `Row` of the animals table and computes whether the animal is a dog.

Transformer (check one) ☐ Filter ☐ Transform ☐ Build

filter-is-dog

Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table

name	species
sasha	cat
fritz	dog
toggle	dog

Transformed Table

name	species
fritz	dog
toggle	dog

Contents (Contract, Purpose Statement, and Expression) ☐

Row -> *Boolean*

Domain *Range*

consumes an animal and computes whether the species is "dog"

Purpose: what does the formula do for each row?

species = "dog"

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Design Recipe 2

Directions: Create and save a Transformer that produces a copy of the animals table with only animals that are more than 12 years old.

Transformer (check one) [x] Filter [ ] Transform [ ] Build

filter-is-old

Transformer name

Example Tables

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table

name	age (years)
Sasha	1
Felix	16

Transformed Table

name	age (years)
Felix	16

Contents (Contract, Purpose Statement, and Expression)

Row -> *Boolean*

Domain *Range*

Consumes an animal, and computes whether its age is > 12

Purpose: what does the formula do for each row?

Age > 12

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Directions: Create and save a Transformer which returns true in a *new column* if an animal's name contains the letter `s`.

Transformer (check one) [ ] Filter [ ] Transform [ ] Build

build-s-names

Transformer name

Example Tables

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table

name	species
Sasha	cat
Felix	cat

Transformed Table

name	species	Sname
Sasha	cat	true
Felix	cat	false

Contents (Contract, Purpose Statement, and Expression)

Row -> *Boolean*

Domain *Range*

Consumes an animal, and computes whether its name contains an `s`

Purpose: what does the formula do for each row?

includes(name,"s")

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Chaining

You have created and saved the following Transformers. (Read them *carefully!*)

`filter-is-fixed`	`filter-is-young`	`order-by-sex`
Transformer: Filter Expression: Fixed = true	Transformer: Filter Expression: Age < 4	Transformer: Sort Attribute: Sex Direction: descending
`order-by-age`	**`order-by-pounds`**	**`order-by-sex`**
Transformer: Sort Attribute: Age Direction: ascending	Transformer: Sort Attribute: Pounds Direction: descending	Transformer: Sort Attribute: Sex Direction: descending

The table below represents four animals from the shelter:

name	sex	age	fixed	pounds
`"Toggle"`	`"female"`	`3`	`true`	`48`
`"Fritz"`	`"male"`	`4`	`true`	`92`
`"Nori"`	`"female"`	`6`	`true`	`35.3`
`"Maple"`	`"female"`	`3`	`true`	`51.6`

What table do we when we apply the indicated Transformers? Match the instructions on the left with the outcome on the right.

Apply `order-by-age`	**1** -*C*	**A**	Produces a table containing only Toggle and Maple
Apply `filter-is-fixed`	**2** -*E*	**B**	Produces a table of only young, fixed animals
Apply `filter-is-young`	**3** -*D*	**C**	Produces a table, sorted youngest to oldest
First, apply `filter-is-young` Next, apply `filter-is-fixed`	**4** -*A*	**D**	Produces a table containing Maple and Toggle, in that order
First, apply `filter-is-young` Next, apply `order-by-pounds`	**5** -*D*	**E**	Produces a table containing the same four animals
Apply `order-by-pounds`	**6** -*D*	**F**	Produces a table of all four animals, with Fritz first

Chaining 2: Order Matters

You have created and saved the following Transformers. (Read them *carefully!*)

`build-kilos`	`order-by-kilos`	`filter-is-female`
Transformer: Build Attribute Name of new attribute: `kilos` Expression: pounds/2.2	Transformer: Sort Attribute: `kilos` Direction: ascending	Transformer: Filter Expression: `sex = "female"`
`filter-is-heavy`	**`order-by-sex`**	**`build-is-female`**
Transformer: Filter Expression: `kilos > 25`	Transformer: Sort Attribute: `sex` Direction: ascending	Transformer: Build Attribute Name of new attribute: `is-female` Expression: `sex = "female"`

The table below represents four animals from the shelter:

name	sex	age	fixed	pounds
`"Toggle"`	`"female"`	3	`true`	48
`"Fritz"`	`"male"`	4	`true`	92
`"Nori"`	`"female"`	6	`true`	35.3
`"Maple"`	`"female"`	3	`true`	51.6

Match the CODAP instructions (left) to what they do (right). **Note: one description might match multiple expressions!**

Apply `order-by-kilos`	**1** -*D*	**A**	Produces a table containing Toggle, Nori and Maple, with an extra column showing their weight in kilograms
Apply `filter-is-female` Next, apply `build-kilos`	**2** -*A*	**B**	Produces a table containing Maple, Nori and Toggle (in that order)
Apply `build-kilos` Next, apply `filter-is-heavy`	**3** -*C*	**C**	Produces a table containing only Fritz.
First, apply `filter-is-heavy` Next, apply `build-kilos`	**4** -*D*	**D**	Won't run: will produce an error
First, apply `build-kilos` Next, apply `filter-is-heavy` Next, apply `order-by-sex`	**5** -*C*	**E**	Produces a table containing only Fritz, with two extra columns.
Apply `build-is-female` Apply `build-kilos` Apply `filter-is-heavy`	**6** -*E*	**F**	Produces a table containing Maple and Fritz

What's on your mind?

Probability, Inference, and Sample Size

How can you tell if a coin is fair, or designed to cheat you? Statisticians know that a fair coin should turn up "heads" about as often as "tails", so they begin with the **null hypothesis:** they assume the coin is fair, and start flipping it over and over to record the results.

A coin that comes up "heads" three times in a row could still be fair! The odds are 1-in-8, so it's totally possible that the null hypothesis is still true. But what if it comes up "heads" five times in a row? Ten times?

Eventually, the chances of the coin being fair get smaller and smaller, and a Data Scientist can say "this coin is a cheat! The chances of it being fair are one in a million!"

By sampling the flips of a coin, we can *infer* whether the coin itself is fair or not. **Inference** plays a major role in Data Science and Statistics!

- If we survey pet owners about whether they prefer cats or dogs, the **null hypothesis** is that the odds of someone preferring dogs are about the same as them preferring cats. And if the first three people we ask vote for dogs (a 1-in-8 chance), the null hypothesis could still be true! But after five people? Ten?
- If we're looking for gender bias in hiring, we might start with the null hypothesis that no such bias exists. If the first three people hired are all men, that doesn't necessarily mean there's a bias! But if 30 out of 35 hires are male, this is evidence that undermines the null hypothesis and suggests a real problem.
- If we poll voters for the next election, the **null hypothesis** is that the odds of voting for one candidate are the same as voting for the other. But if 80 out of 100 people say they'll vote for the same candidate, we might reject the null hypothesis and infer that the population as a whole is biased towards that candidate!

Sample size matters! The more bias there is, the smaller the sample we need to detect it. Major biases might need only a small sample, but subtle ones might need a huge sample to be found. However, choosing a **good sample** can be tricky!

Random Samples are a subset of a population in which each member of the subset has an equal chance of being chosen. A random sample is intended to be a representative subset of the population. The larger the random sample, the more closely it will represent the population and the better our inferences about the population will tend to be.

Grouped Samples are a subset of a population in which each member of the subset was chosen for a specific reason. For example, we might want to look at the difference in trends between two groups ("Is the age of a dog a bigger factor in adoption time v. the age of a cat?"). This would require making grouped samples of *just the dogs* and *just the cats*.

Finding the Trick Coin

There are three coins defined in the [Fair Coins Starter File](): `coin1`, `coin2`, and `coin3`.

One of these coins is fair, one will land on "heads" 75% of the time, and one will land on "heads" 90% of the time. ***Which one is which?***

1) In the table below, record five flips of each coin and the number of "heads" you saw. Then convert that number into a *percentage*. Finally, write a "Y" or "N" depending on whether you think each coin is fair based on your sample.

Sample	coin1	coin2	coin3
1			
2			
3			
4			
5			
#heads	/5	/5	/5
% heads	%	%	%
fair?			

2) In the table below, record 15 more flips of each coin and the number of "heads" you saw. Then convert that number into a *percentage*. Finally, write a "Y" or "N" depending on whether you think each coin is fair based on your sample.

Sample	coin1	coin2	coin3
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
#heads	/20	/20	/20
% heads	%	%	%
fair?			

3) Which weighted coin did you identify first: fair, 75%, or 90%? _____

4) Why was that coin the easiest to find? *The more the outcome matches the null hypothesis, the larger the sample you'll need to detect it.*

Sampling and Inference

1) In the screenshots of the "Sampler" (below), show how you would create a small random sample of 10 animals and a large random sample of 40 animals. *To create two separate tables (rather than a single hierarchical table), re-select and re-open "Sampler" from the Plugins menu before each sampling simulation.*

2) In the options tab, did you select "with replacement" or "without replacement"? Why?

_____

3) Make a bar chart for the animals in each sample, showing percentages of fixed and unfixed.

- The percentage of fixed animals in the entire population is: **47.7%**
- The percentage of fixed animals in the small sample is: _____
- The percentage of fixed animals in the large sample is: _____

4) Make a bar chart for the animals in each sample, showing percentages for each species.

- The percentage of tarantulas in the entire population is: **roughly 5%**
- The percentage of tarantulas in the small sample is: _____
- The percentage of tarantulas in the large sample is: _____

5) Direct the sampler to generate a different set of random samples of these sizes. Make a new bar chart for each sample, showing percentages for each species.

- The percentage of tarantulas in the entire population is: **roughly 5%**
- The percentage of tarantulas in the small sample is: _____
- The percentage of tarantulas in the large sample is: _____

6) Which repeated sample gave us a more accurate inference about the whole population? Why?

_____

_____

Grouped Samples from the Animals Dataset

You've already created and saved the following transformers: `filter-is-old`, `filter-is-young`, `filter-is-cat`, `filter-is-dog`, `filter-is-female`, `filter-is-fixed`, and `filter-has-s-name`. Provide the transformers you would use in the order you would use them. We've given you the solution for the first sample, to get you started.

Students answers will vary the order of the methods chained

	Subset	List the transformers *in order*	Use function notation
1	Kittens	`filter-is-young`, `filter-is-cat`	filter-is-young(filter-is-cat(animals-table))
2	Puppies	`filter-is-dog`, `filter-is-young`	*filter-is-dog(filter-is-young(animals-table))*
3	Fixed Cats	`filter-is-fixed`, `filter-is-cat`	*filter-is-fixed(filter-is-cat(animals-table))*
4	Cats with "s" in their name	`filter-has-s-name`, `filter-is-cat`	*filter-has-s-name(filter-is-cat(animals-table))*
5	Old Dogs	`filter-is-old`, `filter-is-dog`	*filter-is-old(filter-is-dog(animals-table))*
6	Fixed Animals	`filter-is-fixed`	*(filter-is-fixed(animals-table))*
7	Old Female Cats	`filter-is-female`, `filter-is-cat`, `filter-is-old`	*filter-is-female(filter-is-cat(filter-is-old(animals-table)))*
8	Fixed Kittens	`filter-is-fixed`, `filter-is-cat`, `filter-is-young`	*filter-is-fixed(filter-is-cat(filter-is-young(animals-table)))*
9	Fixed Female Dogs	`filter-is-fixed`, `filter-is-female`, `filter-is-dog`	*filter-is-fixed(filter-is-female(filter-is-dog(animals-table)))*
10	Old Fixed Female Cats	`filter-is-old`, `filter-is-fixed`, `filter-is-female`, `filter-is-cat`	*filter-is-old(filter-is-fixed(filter-is-female(filter-is-cat(animals-table))))*

Displaying Data

Fill in the tables below, then use Pyret to make the following displays. Record the code you used. The first table has been filled in for you.

1) A `bar-chart` showing how many puppies are fixed or not.

What Rows?	Which Column(s)?	What Display?
puppies	*fixed*	*bar-chart*

2) A `bar-chart` showing how many heavy dogs are fixed or not.

What Rows?	Which Column(s)?	What Display?
heavy dogs	*fixed*	*bar-chart*

3) A `box-plot` of the number of `pounds` that kittens weigh.

What Rows?	Which Column(s)?	What Display?
kittens	*pounds*	*box-plot*

4) Describe **your own grouped sample** here, and fill in the table below.

What Rows?	Which Column(s)?	What Display?

code: Student answers will vary

Data Cycle: Analyzing Categorical Data

Ask Questions

How many of each species are fixed at the shelter?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, write an expression for your Filter Transformer here.

If you need to Transform or Build an attribute, write the expression for your Transformer here.

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

Ask Questions

Are there more female cats than male cats at the shelter?

What question do you have?

What type of question is this? (circle one) Lookup Arithmetic Statistical

Consider Data

What Rows should we investigate? (All the rows, just the cats, fixed dogs, etc.)

What Columns do we need? (age, weight-in-kilograms, weeks, etc.)

Analyze Data

If you only need some rows, write an expression for your Filter Transformer here.

If you need to Transform or Build an attribute, write the expression for your Transformer here.

Interpret Data

What did you find out? What can you infer?

What new question(s) does this raise?

What's on your mind?

Design Recipe

Directions:

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table	

Transformed Table	

Contents (Contract, Purpose Statement, and Expression) ☐

Row ->

Domain *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Directions:

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

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Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

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Transformed Table	

Contents (Contract, Purpose Statement, and Expression) ☐

Row ->

Domain *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.

Directions:

Transformer (check one) ☐ **Filter** ☐ **Transform** ☐ **Build**

Transformer name

Example Tables ☐

What gets filtered/transformed/built? In the sample tables below, (if needed) add the relevant columns.

Original Table	

Transformed Table	

Contents (Contract, Purpose Statement, and Expression) ☐

Row ->

Domain *Range*

Purpose: what does the formula do for each row?

i.e. Weight < 20 or Species = "rabbit". Pay careful attention to capitalization and quotation marks.